Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/29/2025
PI Name:	Iyer, Janani Ph.D.		
Project Title:	Neurological, Cardiac, and Muscular Consequences of the Deep Space Environment in Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	16
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2022 Space Biology NNH22ZDA001N-SBBLEO E.11: Research Pathfinder
Start Date:	11/01/2023	**End Date:**	10/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	No changes		
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Performance Goal No.:			
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Task Description:	August 2025: Updated Task Description per the Principal Investigator (PI) Human space exploration and long-duration habitation in space necessitate a thorough understanding of the impact of the spaceflight environment on human physiology. This study aims to advance NASA's efforts by swiftly characterizing risks and identifying suitable countermeasures for future deep space missions. We hypothesize that exposure to spaceflight conditions, including altered gravity and radiation, induces oxidative stress responses, leading to behavioral and mitochondrial impairments, morphological changes in neural tissues, heart and muscle, and alterations in gene expression. Additionally, we propose that overexpression of an antioxidant gene, such as superoxide dismutase (SOD2), may mitigate these changes. This multi-system study will utilize Drosophila melanogaster to investigate the prolonged effects of spaceflight and post-flight acclimation. The fly model has exceptional genetic tools, requires limited laboratory maintenance, is well-established as a spaceflight model organism, and has been successfully flown by the Principal Investigator team to LEO using the proposed VFB hardware. This investigation will use various assays to understand the molecular, morphological, functional, and behavioral alterations induced by space travel. Using different genetic fly lines, genetic, cellular, and functional changes will be characterized in space-reared flies and flies readapted to Earth conditions (R+10 days). This study will (a) focus on the cellular and molecular responses to oxidative stress and autophagy caused by microgravity and radiation in muscle, heart, and nervous system, (b) utilize fly lines with tissue-specific expression (neuronal, muscle, and cardiac) of an antioxidant gene, SOD (superoxide dismutase), and (c) use the well-characterized genetic mutant, Tor7/P, to test whether increased autophagy is beneficial or represents a maladaptive response to the spaceflight stressors. Data from this mission will also be compared with previous LEO-based data to identify shared signatures. Furthermore, cross-species analysis of the transcriptomic data will be performed to compare this mission to other invertebrate and vertebrate spaceflight-related data to determine the evolutionarily conserved pathways perturbed by space stressors across species. This project aims to provide crucial insights into the mechanisms underlying oxidative stress responses, synaptic changes, and heart and muscle deficits, facilitating the identification of diagnostic and therapeutic targets to mitigate the adverse health effects of long-duration space habitation. Ultimately, this research will contribute to our ability to thrive in deep space and inform future missions to the Moon and Mars.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	August 2025 Update per PI Potential Impact: The experiments proposed here will test the hypothesis that oxidative stress induced by spaceflight stressors negatively impacts the CNS, heart, and somatic muscles and that tissue-specific overexpression of an antioxidant gene such as superoxide dismutase, can mitigate these effects. This study will use different genetic lines to characterize the genetic, cellular, and functional changes in space-reared flies and their adaptation to Earth's environment across multiple organ systems. We predict sex-specific differences in neuronal, cardiac, and muscular outcomes, informing distinct male-female responses to spaceflight exposure. Detailed behavioral, structural, and multi-omics data from brain, muscle, and heart tissue obtained from the same animals -- permitting an integrated assessment of the effects of the space environment across tissues.
Task Progress:	October 2025 Update: The study has been deprioritized and the flight investigation aboard the International Space Station (ISS) was cancelled. No further reporting is expected (Ed., 10/29/25). To explore the effects of spaceflight and Earth readaptation on the neural, cardiac, and somatic muscle of Drosophila (Aims 1 and 2), we plan to send several Drosophila strains to the ISS: fly lines with tissue-specific expression (neural, muscle, and cardiac) of the antioxidant gene superoxide dismutase 2, (SOD2), and a well-characterized genetic mutant, Tor7/P, to test whether increased expression of an antioxidant gene (SOD2) or autophagy is beneficial or represent maladaptive responses to spaceflight stressors. In the past year, we: • Completed the Experiment Verification Test Readiness Review (EVTRR) • Conducted the Experiment Verification Test (EVT) at Ames Post-analysis of EVT results, we conducted more tests: • Recalibrated the incubator • Tested 2 different fly food recipes - Mitomunch and Revised Mitomunch • Science tests: o Egg lay scenario with nominal (31 days) on Mitomunch and Revised Mitomunch with two different initial fly seedings. o Adult loading scenario with nominal (31 days) on Mitomunch and Revised Mitomunch with two different initial fly seedings. o Due to significant changes in the mission timeline and crew composition, and following consultation with Space Biology management, the R+10 time point was removed. o Extensive testing confirmed that the egg-lay strategy using Mitomunch food is the optimal approach. o The conclusion aligned with the initial Science Verification Test (SVT) testing conducted before EVT-1. We believe the EVT failure was caused by excessive humidity in the incubator used during EVT testing.
Bibliography Type:	Description: (Last Updated: 04/08/2025)